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SilicaFlux semiconductor pre-fab strategy: a silica-dominant wafer platform for the SystemVerilog architecture

Executive summary

The most important conclusion is that **a literally pure-SiO_2 semiconductor cannot implement SilicaFlux as a conventional digital CPU/SoC**. Stoichiometric fused silica is an amorphous silicon-dioxide dielectric, not a useful transistor channel material; SiO_2 has a very large bandgap of roughly 9 eV and is used precisely because it is an excellent electrical insulator and gate dielectric. It can form resistive-switching structures only after introducing defects, oxygen deficiency or conductive filaments, at which point the material is no longer pristine “pure silica”. [1]

The technically credible interpretation of a **“pure-silica-based” SilicaFlux process** is therefore a *silica-dominant heterogeneous platform*: use high-purity fused silica as the mechanical/insulating handle wafer, transfer a thin single-crystal silicon layer onto it for MOSFETs and SRAM, retain SiO_2 as the dominant dielectric and isolation material, and optionally add Si/SiN photonics and SiO_x resistive memory in later experimental revisions. Single-crystal silicon films have been transferred onto glass by ion-cut/Smart-Cut-type processes, while poly-Si TFTs have also been demonstrated directly on fused silica; those approaches provide a scientific basis for a silicon-on-glass or silicon-on-fused-silica platform. [2]

For **first silicon**, however, I would not begin with a bespoke fused-silica-handle process. The lowest-risk route is a **130 nm-class SOI or mature CMOS process**, then migrate to a fused-silica handle after the RTL, SRAM macros, physical design, DFT and package have been validated. Mature 130 nm design rules already support roughly 140–150 nm poly/metal critical dimensions, multiple metal layers, SRAM structures and 1.8 V core devices, and MPW programmes remain available at mature nodes. [3]

This recommendation fits SilicaFlux unusually well. Direct inspection of the supplied archive shows that its primary architecture is a 32-bit scalar/dataflow machine with a SIMD vector engine, a 4×4 tensor unit, four compute tiles, local SRAM, a 2×2 mesh NoC, a register bank, a scheduler and chiplet-oriented physical interfaces. The repository also contains auxiliary RISC-V, pipelined-CPU, cache, DSP, communications and formal-verification libraries. The supplied technical review independently describes approximately 30,768 lines across 179 RTL/test files, 176 modules, 284 assertions and 43 testbench files, while noting that verification is not yet at production UVM/coverage-closure level. 🗄️filecite🗄️turn0file0🗄️

The primary SilicaFlux synthesis constraints in the uploaded source target **100 MHz** on FPGA. For ASIC planning, I would therefore use three design points: a low-risk 100–200 MHz implementation at 130/180 nm; a medium target around 250–500 MHz using an optimised 90/65 nm or FD-SOI implementation; and a high-performance 500 MHz–1+ GHz target only after substantial physical retiming, SRAM macro substitution, clock-tree engineering and probably migration to 28 nm-class FD-SOI or below. These are engineering targets rather than guarantees; the actual frequency cannot be established until logic synthesis, SRAM characterisation, extraction and static timing analysis are run against a real PDK.

My recommended pre-fab is therefore:

fused-silica handle → bonded 100–250 nm single-crystal Si device film → 130 nm-class planar CMOS → 4–6 metal BEOL → SiO_2-rich interlayer dielectrics → optional Si/SiN photonics → optional SiO_x RRAM test array → Cu redistribution / flip-chip package.

That gives SilicaFlux the electrical behaviour of a conventional CMOS SoC while making silica the dominant substrate and dielectric material. It is substantially more credible than trying to fabricate the CPU itself from SiO_2.

Material and device feasibility

Why pure silica cannot be the transistor material

Fused silica is synthetic amorphous SiO_2. Commercial high-purity fused silica has a thermal conductivity of only about 1.3 W m^{-1} K^{-1} and a coefficient of thermal expansion around 0.5 ppm K^{-1}; these properties make it attractive as an electrical/optical substrate but poor as a heat spreader. [4] Silicon dioxide's wide bandgap and high breakdown strength are why it historically served as the MOS gate dielectric rather than the conducting channel. [5]

Consequently, a SilicaFlux CPU fabricated solely from stoichiometric SiO_2 would have no practical CMOS transistor network. The nearest "silica electronics" options all add another electronic phase: crystalline or polycrystalline silicon for FETs, silicon-rich SiO_x for resistive memory, Si or SiN for photonics, or III–V/Ge materials for specialised optical or tunnelling devices. This distinction should be treated as a fundamental process constraint, not merely a fabrication inconvenience. [6]

The most viable material choices

A **single-crystal silicon-on-fused-silica** structure is the closest practical match to the requested concept. Ion-cut and Smart-Cut approaches have transferred single-crystal Si onto glass, and the resulting semiconductor film can support conventional MOS devices while the bulk substrate remains insulating glass. [7] The more commercially mature analogue is SOI, in which a silicon device layer sits on a buried SiO_2 layer; SOI device layers can range from only a few nanometres to micrometre scale and BOX thickness can range from nanometres to micrometres depending on the application. [8]

For SilicaFlux's first ASIC generation, **MOSFETs should remain the default device**. The CPU pipeline, scheduler, NoC routers, vector lanes, tensor multipliers and control logic are all conventional synchronous digital structures and map naturally to standard-cell CMOS. SRAM should likewise use foundry-qualified 6T or 8T bit cells rather than experimental oxide memory. Mature 130 nm PDKs already expose standard digital cells, SRAM structures and I/O devices. [9]

TFETs are not recommended for the first SilicaFlux tape-out. Tunnel FETs can achieve sub-60 mV/decade subthreshold slopes, but published studies continue to report drive-current, integration and high-performance trade-offs relative to MOSFET technology. Hybrid TFET/MOSFET technology is scientifically credible but process-complex and not needed to meet the current 100 MHz-class architectural baseline. [10]

SiO_x memristors are appropriate only as an optional research memory/accelerator macro. Resistive switching has been repeatedly observed in oxygen-deficient silicon oxide and is associated with oxygen vacancies, Si-rich filaments or related defect-mediated conduction. That means SiO_x RRAM

is interesting for weights, nonvolatile configuration or future compute-in-memory, but it should not replace conventional SRAM in the first version of the cache, register file or tile memories. [11]

Silicon photonics is a realistic later addition, particularly for the chiplet-oriented interfaces already present in SilicaFlux. Commercial/research platforms use SOI structures such as approximately 220 nm Si on a 2 µm buried oxide and integrate waveguides, modulators, photodetectors and packaging structures on 200 or 300 mm wafers. [12] This would not make the processor “optical”; it would replace high-bandwidth die-to-die electrical links with optical I/O where bandwidth-per-edge or package reach justifies the cost.

Recommended performance envelopes

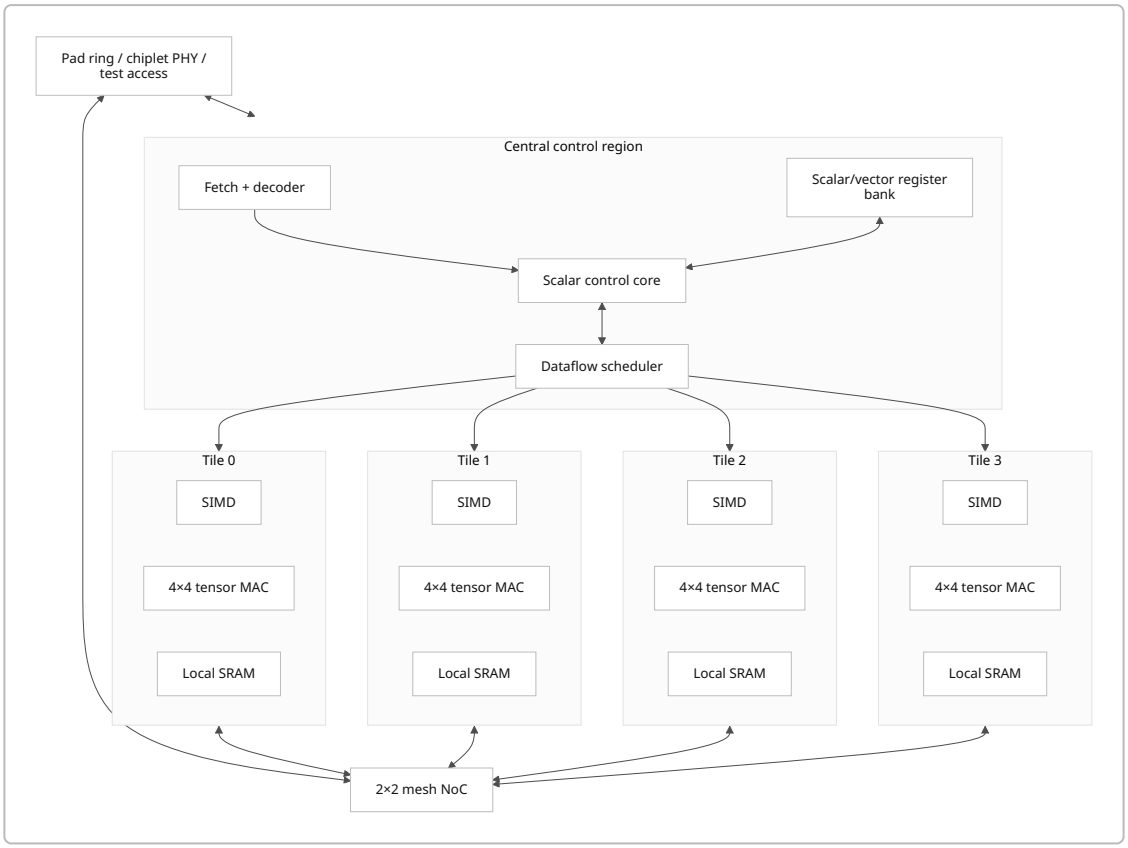
SilicaFlux target	Recommended device/ process	Nominal core frequency target	Core voltage class	Design priority
Low-risk prototype	130–180 nm planar CMOS, preferably SOI or silicon-on-glass experimental derivative	100–200 MHz	~1.2–1.8 V depending PDK	correctness, yield, DFT
Balanced ASIC	65–90 nm CMOS/SOI	250–500 MHz	~0.9–1.2 V	energy/ throughput
High-performance	22–28 nm FD-SOI or comparable advanced CMOS	500 MHz–1+ GHz	~0.7–1.0 V	PPA, CTS, SRAM optimisation
Silica research variant	130–250 nm silicon-on-fused-silica custom process	initially ≤100–200 MHz	process-dependent	substrate/ material validation

The mature-node baseline is consistent with documented 130 nm-class rules: SKY130, for example, has field-oxide dimensions around 0.14 µm, poly dimensions around 0.15 µm and first-metal dimensions around 0.14 µm, with local interconnect and multiple metal layers. These numbers are useful **reference targets only**; a custom silicon-on-fused-silica process would need its own complete PDK and statistical process qualification. [13]

SilicaFlux physical mapping and floorplan

The architecture should be floorplanned as a **compute-centric four-quadrant die**. The four tiles are natural repeated physical blocks; each should contain its vector engine, tensor engine and local SRAM, with the four NoC routers placed near the tile-facing corners. The scalar control core, fetch/decode logic, register bank and scheduler should occupy a compact central band so that control, barrier and register traffic does not traverse the entire die. This follows the actual repository's four-tile/2×2-mesh structure and reduces long global buses.

The supplied report's page-6 SoC diagram similarly places a CPU behind a shared fabric with SRAM/ ROM and peripherals, reinforcing the separation between compute, memory and peripheral regions. filecite turn0file0



The CPU/control region should use a conventional high-density standard-cell library with a mixture of regular-VT and high-VT cells. The tensor and vector blocks should use higher-drive cells on arithmetic carry chains and clock-gated register banks; the multipliers should be synthesised using structured Booth/Wallace/CSA implementations or foundry multiplier macros where available. The local SRAMs should be hard macros rather than flip-flop arrays. Clock-gating cells should be limited to qualified integrated clock-gating cells rather than RTL-generated logic gates.

SilicaFlux module/ IP	Suggested physical implementation	Preferred device/library	Key process constraint
Fetch/decode/ control core	dense standard-cell region	CMOS RVT/HVT mix	short control paths; low skew
Scalar ALU	placed beside control core	CMOS arithmetic cells	carry-chain timing
Register bank	latch/FF bank initially; compiled RF macro later	high-density CMOS	read/write port congestion
Vector engine	repeated SIMD lane rows	CMOS + structured multiplier cells	high local routing density
4×4 tensor unit	regular MAC array	CMOS Booth/CSA tree	switching power and IR drop
Tile SRAM	compiled SRAM macro	6T/8T CMOS SRAM	retention, bit-line noise
Scheduler	central standard cells	CMOS RVT	fan-out to four tiles

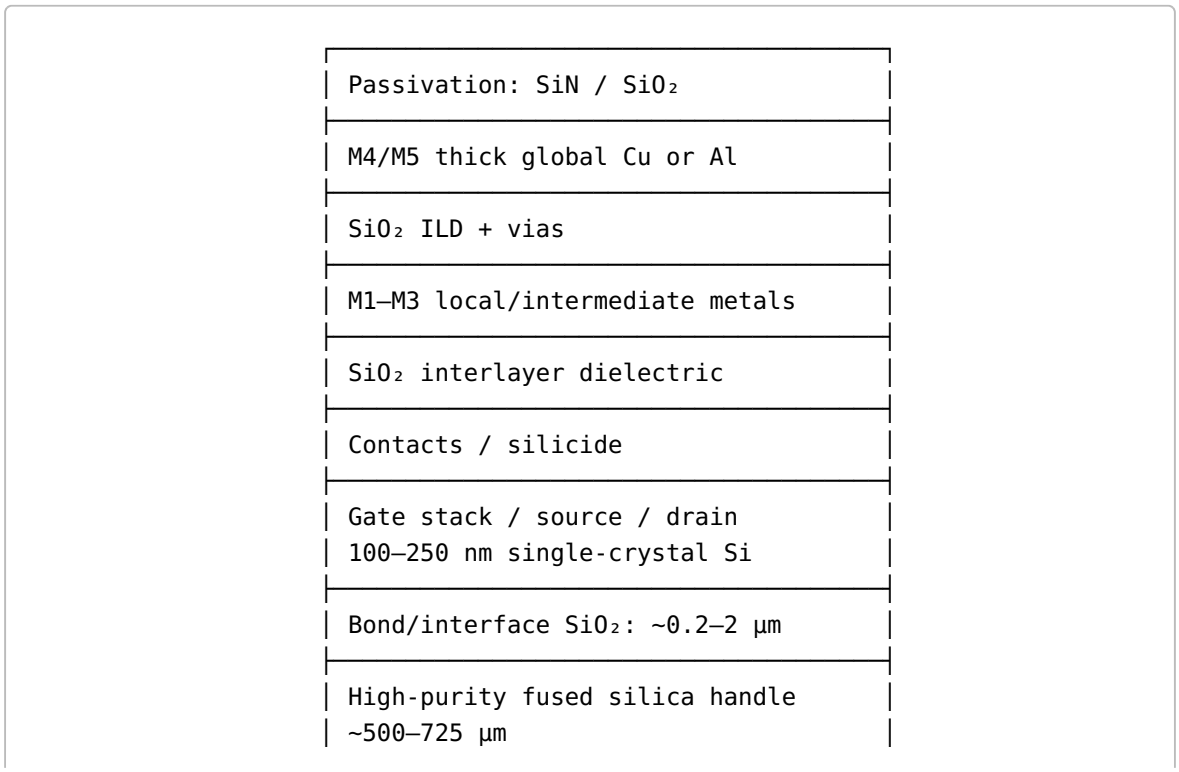
SilicaFlux module/ IP	Suggested physical implementation	Preferred device/library	Key process constraint
2×2 NoC	central cross-shaped routing channels	CMOS + upper-metal global wires	congestion and latency
Memory/cache library	SRAM macros + tag logic	CMOS SRAM + standard cells	DFT/BIST mandatory
UART/SPI/I^2C/ GPIO	pad-ring peripheral region	thick-oxide I/O CMOS	ESD and voltage-domain rules
Chiplet interface	die-edge PHY	CMOS serializers; optional photonics later	SI, ESD, clock-domain crossing
Experimental NVM	isolated test macro	SiO_x RRAM	forming variability/ endurance
Future optical link	edge photonic block	Si/SiN waveguides + Ge PD where available	separate photonics PDK

The upper two metal layers should be reserved largely for VDD/VSS distribution, clock trunks and long NoC links. At a 130 nm-class reference process, the existence of five metal levels plus local interconnect is already a realistic template. [14]

Recommended silica-dominant pre-fab process

Wafer stack

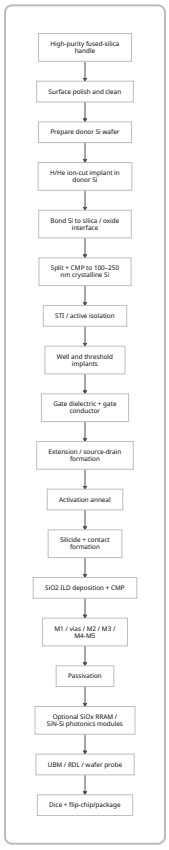
The preferred research wafer is not "SiO_2 only"; it is a **thin semiconductor layer on a silica handle**.



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                    ├──────────────────────────────────────┤
                    │ optional backside Cu thermal/RDL     │
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For comparison, established silicon-photonics SOI platforms commonly use about 220 nm Si above a 2 µm BOX, demonstrating that a very thin crystalline Si active layer over a thick silica dielectric is manufacturable at wafer scale. [15]

Proposed fabrication flow



The following recipe is a **process-development baseline**, not a qualified production run sheet. Exact implant doses, gas flows, etch chemistries and anneal windows must be established by TCAD, split-lot experiments and electrical PCM results.

Process stage	Recommended starting point	Rationale
Handle wafer	200 mm fused silica, ~500–725 µm	compatible with research tooling and strong electrical isolation
Transferred Si	100–250 nm single crystal	enough body thickness for relaxed 130 nm planar CMOS
Bond oxide/ interface	~0.2–2 µm SiO_2	isolation and bonding margin
Active CD target	150–250 nm first generation	less aggressive than leading 130 nm rules to improve custom-process yield

Process stage	Recommended starting point	Rationale
STI	~250–400 nm trench depth	device isolation
Gate oxide	~3–6 nm equivalent SiO_2 for core; thicker I/O option	mature low-voltage CMOS baseline
Gate conductor	poly-Si for simplest mature process, or metal gate if line supports it	poly-Si lowers custom process complexity
S/D junction depth	order 50–120 nm	comparable to mature-node shallow-junction scale; SKY130 reference is ~0.1 µm [13]
Interlayer dielectric	PECVD/HDPCVD SiO_2, ~0.5–1 µm per level	keeps stack silica-rich
Local metals	Al or Cu, ~0.3–0.8 µm	mature BEOL
Top power metal	~1–3 µm Cu/Al	reduce IR drop
Passivation	~0.5–1 µm SiN over SiO_2	moisture/ionic protection
UBM	Ti/TiW barrier + Cu/Ni/Au or equivalent qualified stack	flip-chip/package interface

The front-end thermal budget deserves special attention. Fused silica itself has a high softening temperature, but the silicon/silica bonded interface and the strong difference between silica and silicon thermal expansion can generate stress during high-temperature cycles. Commercial fused silica has a CTE around 0.5 ppm/K, so a custom process should minimise prolonged high-temperature cycles after bonding and use stress-monitor structures across the wafer. [16]

For this reason, the most conservative integration strategy is to **fabricate much of the CMOS device stack on the silicon donor first, then transfer/bond**, or use a reduced-temperature post-bond process. Literature on circuit transfer explicitly describes bonding processed silicon to foreign substrates and thinning the silicon afterwards. [17]

Mask strategy

A first silicon-on-silica SilicaFlux process would likely require roughly **20–30 design masks** even before optional photonics or RRAM. A mature 130 nm reference technology typically has many more process options and can run around thirty-plus mask steps, illustrating why a custom wafer process becomes expensive quickly. [18]

The essential mask families are active/STI; wells and threshold-adjust implants; gate; N/P extension and source/drain; contact; each interconnect and via level; passivation/pad opening; and optional RDL/UBM. Dedicated RRAM masks would add bottom electrode, SiO_x switching layer definition and top electrode. Photonics would add shallow/deep Si etches, SiN waveguides, modulator implants and optical opening/edge-coupler layers. Foundry photonics PDKs already package comparable component and verification layers. [19]

DFM, DFT, wafer test and packaging

The current SilicaFlux codebase should **not go directly from RTL to a custom silica process**. The supplied review found 284 assertions but no production UVM infrastructure and only relatively basic testbenches. filecitefileturn0file0 A costly custom wafer run should be gated by lint closure, CDC/RDC closure, synthesis, gate-level simulation with SDF, formal checking of protocol/state-machine properties, functional coverage, equivalence checking and full DFT insertion.

DFT should include scan on all synthesised sequential logic, separate SRAM MBIST for each tile SRAM and cache macro, boundary scan on external I/O, JTAG or equivalent test access, and on-chip clock control for at-speed scan. The repeated four-tile structure should also support tile-level isolation so one defective accelerator tile can be characterised independently. The NoC should have loopback and packet-BIST modes.

A custom wafer needs extensive **process-control monitors** in the scribe lanes: NMOS/PMOS transistor arrays at several geometries, Van der Pauw structures, contact/via chains, serpentine metal opens/bridges, ring oscillators, SRAM bitcells, MOS capacitors, oxide-breakdown structures, cross-bridge Kelvin resistors and diode leakage structures. Silicon-photonics lines similarly rely on wafer-scale process-control testing before individual dies are released. [15]

The fused-silica handle substantially changes thermal design. Fused silica conducts heat at only about 1.3 W m^{-1} K^{-1}, far below silicon, so a tensor-heavy SilicaFlux die will self-heat more strongly than the same active silicon on a conventional bulk-silicon handle. [20] The package should therefore remove heat primarily through the **front side** using flip-chip bumps, a copper lid/heat spreader and a high-conductivity TIM. Dense VDD/VSS bump arrays should be placed close to tensor/vector tiles rather than relying on lateral power transport through the die.

The preferred package is a modest flip-chip BGA or wafer-level fan-out package. The four tile regions should each receive independent power/ground bump clusters, with the control core on a quieter power island if feasible. The experimental photonic version should reserve one die edge for optical couplers and keep high-current tensor power bumps away from that edge.

Yield, reliability and risk assessment

The largest risk is not transistor lithography; it is **the custom substrate/interface**. Conventional mature CMOS works because decades of process integration have stabilised defect density, implant activation, interfaces and SRAM yield. A silicon-on-fused-silica process introduces wafer bonding defects, film-thickness non-uniformity, thermal stress, interface charge and difficult heat removal. Published SOI work emphasises that device-layer thickness uniformity and BOX reliability directly affect threshold variability and device behaviour. [21]

Risk	Severity	Likelihood	Main mitigation
Pure SiO_2 cannot form CMOS channel	Critical	Certain	use transferred crystalline Si; restrict SiO_2 to substrate/dielectrics
Bond voids / Si-film defects	High	Medium-high	IR bond inspection, CMP metrology, sacrificial monitor wafers

Risk	Severity	Likelihood	Main mitigation
Thermal self-heating	High	High	flip-chip cooling, thick top metal, aggressive power gating
Silicon/silica stress	High	Medium	reduced post-bond thermal budget, wafer curvature monitors
SRAM yield	High	Medium	foundry-qualified macro first; conservative custom bitcell later
Clock/noise failures	Medium-high	Medium	CTS, power-grid modelling, decap insertion, post-layout STA
Tensor IR drop	High	Medium	local bump clusters, power stripes, activity-aware placement
SiO_x RRAM variability	High	High	keep experimental and non-critical in first revision
Photonics integration complexity	Medium-high	Medium	separate chiplet/PIC initially
RTL verification escapes	Critical	Medium	coverage closure, formal, scan/BIST and gate-level validation
Low yield from aggressive CD	High	Medium	begin at ≥150–250 nm custom rules and tighten later

Reliability qualification should cover TDDB of gate/interlayer oxides, bias-temperature instability, hot-carrier effects, electromigration, via stress, SRAM retention, package thermal cycling and ESD. Oxide breakdown is inherently statistical and remains a major reliability consideration in silicon-dioxide films. [22]

Because the fused-silica substrate is thermally resistive, electromigration and temperature acceleration should be modelled using **local junction temperature**, not package ambient. A “high-performance” SilicaFlux on glass without a strong front-side thermal path would be a poor design choice even if transistor timing closed.

Tooling, cleanroom, cost and programme estimate

A 130–250 nm research process does not require EUV, but it still requires a serious semiconductor line. ISO Class 5 is a common semiconductor cleanroom level, and critical lithography/process tools normally operate in locally cleaner mini-environments. [23] Fabrication also introduces substantial chemical and gas hazards: semiconductor processes commonly involve HF, strong oxidisers, silane and dopant precursors such as phosphine, diborane or arsine, so a custom facility requires engineered exhaust, toxic-gas monitoring, interlocks and trained process staff. [24]

Capital item	Role	Rough 200 mm research-line budget, 2026 planning range
ISO 5 cleanroom + utilities	environment, UPW, gases, exhaust	£5m–£20m

Capital item	Role	Rough 200 mm research-line budget, 2026 planning range
i-line/DUV stepper	130–250 nm-class patterning	£2m–£8m used/refurbished
coat/develop tracks	resist processing	£0.5m–£2m
RIE/ICP etchers	Si, oxide, nitride, metal	£1m–£4m
PECVD/LPCVD	SiO_2, SiN, poly-Si	£1m–£4m
oxidation/diffusion furnaces	thermal oxide/anneal	£0.5m–£2m
ion implanter	wells and source/drain	£2m–£8m
sputter/PVD	gate/barrier/metals	£1m–£3m
electroplating	Cu/RDL/bumps	£0.3m–£1m
CMP tools	wafer planarisation	£1m–£3m
wet benches	cleaning/oxide etch	£0.5m–£2m
metrology	CD-SEM, ellipsometry, profilometry	£2m–£8m
wafer prober/test	parametric + digital test	£0.5m–£3m
bonding/thinning equipment	silicon-on-glass fabrication	£1m–£5m

These are **order-of-magnitude planning estimates, not supplier quotations**. A full in-house line would therefore be a tens-of-millions-of-pounds undertaking before staffing, facilities qualification, spares and process-development losses. The far more economical route is a university/research-fab or MPW collaboration. MPW services reduce prototype cost by sharing masks; Europractice states that MPW can reduce the effective cost of a full prototype run to roughly 5–10% of a dedicated run. [25]

For SilicaFlux, a sensible budget hierarchy is:

Programme stage	Rough budget	Calendar duration	Deliverable
RTL-to-GDS on open/mature 130 nm PDK	£50k–£200k engineering/NRE	4–8 months	sign-off GDS, no custom substrate
Conventional MPW first silicon	£20k–£100k fabrication/package/test	3–6 months after tape-out	electrical validation
SOI commercial/research prototype	£100k–£500k	6–12 months	substrate-isolated ASIC
Silicon-on-fused-silica feasibility lot	£0.5m–£2m	12–24 months	transistor/PCM + small test chip

Programme stage	Rough budget	Calendar duration	Deliverable
Full SilicaFlux custom-silica pilot	£2m–£10m+	18–36 months	integrated processor lot
Dedicated pilot facility	£20m–£60m+	several years	process ownership

The first two rows are consistent with the continuing availability of 130 nm MPW routes and the general economics of shared-mask prototyping; the later rows are engineering estimates reflecting the need to develop and qualify a new substrate, transistor and BEOL integration flow rather than simply fabricate an existing PDK. [26]

The raw-material BOM for a custom wafer lot is small compared with lithography and process cost: fused-silica handles, donor silicon wafers, photoresist, high-purity process gases, implant species, CMP slurries, dielectric precursors, Al/Cu/Ti/TiN or related metal targets, passivation materials, probe cards, package substrates and bumps. The expensive “BOM” is effectively mask writing, tool time, cleanroom overhead, metrology and engineering labour.

From an IP perspective, the key separation is between **SilicaFlux RTL IP**, the **standard-cell/SRAM library**, and the **process integration/IP of the custom silicon-on-silica wafer**. An open-source PDK can be used to learn the physical-design flow, but its transistor and SRAM layouts should not be assumed portable to a new substrate. The SkyWater open PDK itself explicitly warns that its public release is experimental rather than a production-qualified substitute for full foundry enablement. [14]

Recommended development path

The optimum strategy is therefore **not to build a bespoke silica fab first**. Freeze a single SilicaFlux RTL generation, because the uploaded archive contains several generations and auxiliary CPU/DSP implementations; then harden that version on a known mature CMOS process, close verification and DFT, fabricate an ordinary MPW chip, and use the measured silicon as the electrical reference for the silica substrate programme. The present codebase is sufficiently substantial to justify this discipline, but the uploaded review's finding of limited coverage-driven verification makes a direct custom-process tape-out financially premature. filecite turn0file0

The second silicon should be an SOI implementation. SOI gives SilicaFlux the electrical isolation, reduced substrate parasitics and oxide-rich structure being sought, without introducing every problem of a fused-silica handle at once. SOI wafer technology is mature at 200 and 300 mm scale, and commercial photonics processes already demonstrate sophisticated devices on thin Si/BOX stacks. [27]

Only after the SOI version is validated should the project move to **silicon-on-fused-silica**. At that stage, I would deliberately keep the transistor rules conservative—approximately 180–250 nm initially—even if the lithography can resolve 130 nm. The purpose of that lot is to validate bonding, transistor characteristics, SRAM retention, thermal behaviour and reliability, not to maximise transistor density.

The final SilicaFlux silica platform would then combine the technologies where they are strongest: crystalline Si MOSFETs for the CPU, vector and tensor engines; conventional SRAM for latency-critical memories; fused silica and SiO_2 for substrate isolation and interlayer dielectrics; SiO_x RRAM only for experimental nonvolatile arrays; and silicon/SiN photonics for future chiplet links. That architecture retains the substantive meaning of a silica-based semiconductor platform while respecting the central

materials fact that **silica itself is the insulator that enables the electronics, not the semiconductor that performs the computation**. [28]

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